

Parasitology in Microbiology: Beginner's Course

Lesson 8: Public Health and Epidemiology

Public Health and Epidemiology – Study Notes

Key Concepts

1. **Transmission cycles** – How pathogens move between hosts, vectors, and the environment.
2. **Outbreak investigation** – Systematic steps to identify the source, cause, and control measures for a disease surge.
3. **Sanitation and environmental health** – The role of clean water, waste disposal, and hygiene in preventing disease.
4. **Epidemiologic measures** – Incidence, prevalence, attack rate, and basic reproduction number (R_0).
5. **Control & prevention strategies** – Vaccination, vector control, health education, and policy interventions.

Summary

Public health focuses on protecting and improving the health of populations, while epidemiology provides the tools to understand how diseases spread. Central to both fields is the concept of **transmission cycles**, which describe the pathways a pathogen follows—from reservoir to host, often involving vectors (e.g., mosquitoes) or environmental media (e.g., contaminated water). Recognizing these cycles helps epidemiologists predict where and how an outbreak may occur.

When an outbreak is detected, investigators follow a structured approach: verify the diagnosis, define cases, describe the outbreak by time, place, and person, develop hypotheses, test them analytically, and implement control measures. Throughout, **sanitation**—including safe water, proper waste disposal, and personal hygiene—remains a foundational preventive measure. Improvements in sanitation have historically reduced the burden of many infectious diseases, illustrating the powerful impact of environmental health interventions on population wellbeing.

Important Points

- **Transmission cycles**
 - **Direct** (person to person, e.g., influenza).
 - **Indirect** via **vectors** (malaria, dengue) or **environment** (cholera, hepatitis A).
 - **Zoonotic** cycles involve animal reservoirs (rabies, Lyme disease).
- **Outbreak investigation steps**
 1. **Confirm** the outbreak (lab tests, case definition).
 2. **Case finding** – active surveillance, line lists.
 3. **Descriptive epidemiology** – epidemic curve, spot map, demographic profile.
 4. **Hypothesis generation** – “What, who, where, when, how?”
 5. **Analytical studies** – cohort or case control to test exposures.

6. **Implement control measures** – isolation, vaccination, source removal.

7. **Communicate findings** – reports, press releases, policy briefs.

- **Sanitation impact**

- Access to **safe drinking water** reduces water borne diseases by >50 % in many settings.

- Proper **fecal waste disposal** interrupts fecal oral transmission cycles.

- Hand washing with soap can lower respiratory and diarrheal infections by 20–30 %.

- **Key epidemiologic metrics**

- **Incidence** = new cases / population at risk (per time period).

- **Prevalence** = existing cases / total population.

- **Attack rate** = (cases in outbreak / exposed population) × /100 %.

- **R** – average secondary cases from one primary case in a fully susceptible population; R > 1 indicates potential for spread.

- **Prevention hierarchy**

1. **Eliminate** the source (e.g., remove contaminated water).

2. **Interrupt** transmission (vector control, quarantine).

3. **Protect** susceptible individuals (vaccines, prophylaxis).

4. **Educate** the public (hygiene promotion).

Examples

1. Cholera Outbreak Linked to Contaminated Water

- **Transmission cycle:** *Vibrio cholerae* lives in water; humans become infected by drinking contaminated water.

- **Investigation:**

- Case definition: acute watery diarrhea ≥ 3 /times/24 /h.

- Epidemic curve showed a point source pattern.

- Spot map identified cases clustering near a broken municipal pipe.

- Water samples confirmed *V. cholerae*.

- **Control:** Immediate chlorination of the water supply, distribution of oral rehydration salts, public education on boiling water, and repair of the pipe.

2. Dengue Fever Surge in an Urban Neighborhood

- **Transmission cycle:** Virus circulates between humans and *Aedes aegypti* mosquitoes (vector borne).

- **Investigation:**

- Cohort study compared households with and without water storage containers.

- Higher attack rate in homes storing water uncovered (mosquito breeding sites).

- **Control:** Source reduction (covering containers, larviciding), community clean up campaigns, and public awareness about eliminating standing water.

Quick Review

1. **What are the three main types of transmission cycles?**

2. **List the seven basic steps of an outbreak investigation.**

3. **How does improving sanitation affect the basic reproduction number (R_0) of a disease?**
4. **Define attack rate and give an example of when it is used.**
5. **Name two control measures that target the vector in a vector borne disease outbreak.**

Further Reading

- **“Principles of Epidemiology in Public Health Practice”** – CDC (free online textbook).
- **Sanitation and Health** – WHO/UNICEF Joint Monitoring Programme reports.
- **Basic Reproduction Number (R_0)** – mathematical foundations and real world applications.
- **One Health Approach** – integrating human, animal, and environmental health.
- **Case Studies in Outbreak Investigation** – classic examples (e.g., 1854 Broad Street cholera, 1995–96 E. coli O157:H7).

